

MC Dropout UQ Report: Baseline vs Multi-input

Test split, n=1,262, 50 Monte Carlo dropout samples, final TensorFlow checkpoints.

Executive takeaway

The multi-input model is better overall: it improves point accuracy, produces much

better calibrated uncertainty, and raises 95% coverage from 75.4% to 90.4%. The cost

is wider prediction intervals: about 42.8 months average width versus 34.8 months for

baseline.

-1.62 mo

MAE improvement

+15.0 pp

PICP@0.95 gain

-76.1%

ECE reduction

+7.93 mo

Headline Metrics

Metric	Baseline	Multi-input	Change	Interpretation
MAE	9.80 mo	8.18 mo	-1.62 mo (-16.5%)	Multi is more accurate on average.
RMSE	12.52 mo	10.77 mo	-1.76 mo (-14.0%)	Multi reduces larger errors too.
PICP@0.95	75.4%	90.4%	+15.0%	Multi intervals cover far more true ages.
MPIW@0.95	34.83 mo	42.76 mo	+7.93 mo (22.8%)	Multi is less sharp but more honest.
ECE central	0.1101	0.0263	-0.0838 (-76.1%)	Multi is much better calibrated.

Coverage vs Nominal Confidence

0%20%40%60%80%100%50%80%90%95%Baseline PICPMulti PICPNominal

PICP should ideally match the nominal confidence. Baseline is consistently below target, especially at high confidence. Multi is much closer, though still slightly under-covering at 90% and 95%.

Interval Width Tradeoff

0 mo9 mo18 mo27 mo36 mo45 mo50%80%90%95%Baseline MPIWMulti MPIW

Multi achieves better coverage partly by expressing larger uncertainty. This is a desirable tradeoff here because baseline intervals are too narrow and miss too often.

Multi preferredCalibrate before deployment

Analysis

1. Accuracy

The multi-input model reduces MAE from 9.80 to 8.18 months and RMSE from 12.52 to 10.77 months. That means the multi model is not only better calibrated, it is also a better point predictor.

2. Coverage

Baseline 95% intervals cover only 75.4% of true labels, so they are overconfident. Multi raises this to 90.4%, which is much closer to the intended 95% confidence level. It is still under-covering at 95%, but the gap falls from 19.6 percentage points to 4.6 percentage points.

3. Sharpness

Baseline has narrower 95% intervals at 34.83 months, while multi uses 42.76 months. Narrower intervals are only valuable when coverage is adequate. In this comparison, baseline is sharp but unreliable; multi is wider but substantially more trustworthy.

4. Calibration

Central ECE drops from 0.1101 to 0.0263, a large improvement. This means the empirical

coverage of multi-input intervals tracks requested confidence levels much more closely

across the calibration curve.

Recommended conclusion

For uncertainty quantification, the multi-input model should be preferred over the baseline. It improves both prediction accuracy and uncertainty calibration. The remaining issue is that even multi-input MC Dropout still under-covers slightly at 95%, so the uncertainty intervals should be presented as model-derived predictive intervals rather than guaranteed confidence intervals unless post-hoc calibration or conformal prediction is added.